Product & Technical Documentation

1. What We Were Asked

Build a lightweight, human-centered appointments product with two sides sharing the same live data:

- **Side A (Client App):** Mobile-first, polished consumer feel. Show the client's next appointment, book a consultation without double bookings, view upcoming/past consultations, and reschedule or cancel.
- **Side B (Nutritionist Dashboard):** List appointments with name, phone, date, time, and status (`Booked`, `Done`, `No-show` in distinct colors). Ability to change status, add an appointment by hand, filter to today's consultations, and load sample data.
- **Part 1 Integration:** Include at least one feature that helps solve the 25% no-show problem.
- **Add what was missed:** Identify missing requirements, build them, and articulate what was added and why.

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2. Your Approach

Rather than building an artificial marketing presentation or a heavyweight admin panel, we designed **Kayro Nourish** as a calm, distraction-free product:

- **Pure CSS Responsiveness:** Designed for 390px mobile viewports first with 16px side padding, 44px minimum tap targets, and bottom-anchored booking CTA. On desktop (`md` and up), the client view centers in a `max-w-md` column, and the nutritionist view expands to a clean `max-w-4xl` table.
- **Persistent Memory & REST Endpoint:** Backed by persistent storage (`localStorage` synchronized with a lightweight REST endpoint at `/api/appointments`), ensuring state persists across page refreshes and server restarts.
- **Slot Transparency & Past-Date Guard:** The slot grid explicitly renders available slots (`Free`) and reserved slots (`Taken`). Past calendar dates and past daily times are locked out.

- **Quiet Routing:** Default route (`/`) is the Client App; `/dashboard` (or `#/dashboard`) is the Nutritionist Dashboard. A small text link in the footer transitions between views.

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3. Decisions and Assumptions (Including Sample Data Handling)

Handling Sample Data

The assignment data contained real-world inconsistencies:

- **Inconsistent date formats:** `24/10/2026` , `2026-10-24` , and `25-10-2026` . We built a normalization parser that standardizes all dates into ISO `YYYY-MM-DD` for chronological sorting while presenting clean readable dates (`Sat, Oct 24, 2026`) in the UI.
- **Inconsistent time formats:** `10:00` , `11:30am` , `4 pm` , `09:00` , `14:00` . Normalized into 24-hour time (`16:00`) for conflict checks and rendered in human 12-hour format (`4:00 PM`).
- **Phone Numbers & Missing Contact Handling (Meera Iyer):** Enforced exactly 10-digit mobile number validation (`^\d{10}$`) on all bookings. In the mock table, `Meera Iyer` had no phone number. In manual WhatsApp workflows, clients without phone numbers are uncontactable. In our app, Meera's record is clearly badged as `No phone on file` , and the reminder action is locked with a prompt to collect contact info upon physical check-in.
- **Duplicate Bookings in Mock Data (Riya Shah):** Riya Shah appeared twice in the original brief (Oct 24 at 10:00 AM and Oct 25 at 4:00 PM), violating the "no double booking" constraint. This is typical of WhatsApp/Excel tracking where a client moves their appointment and the practitioner enters the new slot without removing the old row. In software, we enforce atomic rescheduling and deduplicated the clean test data to eliminate phantom no-shows.
- **Today's Date:** The mock schedule centers around `2026-10-24` as "Today", showing Riya and Arjun's appointments when filtering for today.

Key Product Decisions

1. **Added a 4th Status — `Cancelled` :** The brief specified only `Booked` , `Done` , and `No-show` . If a client cancels, marking them as `No-show` corrupts attendance analytics, and deleting them erases audit history. We added `Cancelled` (slate grey), which preserves the record and immediately frees the time slot.

2. **Defined Working Hours & Slot Intervals:** Established standard operating hours: **09:00–13:00** and **14:00–17:00** in **30-minute intervals**, with a **13:00–14:00 lunch break** and past slots disabled.
3. **Pre-Session Check-in ("Still Coming?") & Reminder Tracking:** Passive reminders don't prevent no-shows. We added an interactive 1-tap confirmation step on the client's next appointment card (`Yes, I'm coming`). On the dashboard, unconfirmed bookings are flagged with an early warning marker (`⚠️ Unconfirmed`), and a dedicated **Reminder** column allows 1-click confirmation pings (`Remind` → `✓ Sent`) with instant toast feedback.
4. **Atomic Rescheduling & 1-Time Rule:** Rescheduling frees the old slot and reserves the new slot in a single transactional step, eliminating any window for duplicate bookings. A 1-time reschedule rule prevents clients from repeatedly pushing bookings without notice.
5. **Duplicate Booking Prevention:** If a client with the same name or phone number attempts to book another session while having an active upcoming booking, the system politely alerts them via toast: *"Already present! You already have an active booking on [Date] at [Time]"*.
6. **Standardized +91 Mobile Formatting:** All phone inputs feature a clean `+91` prefix group, and all phone displays across client views, dashboard tables, and Excel exports consistently format numbers as `+91 XXXXX XXXXX` .
7. **Dual-Mode Visual & JSON API Endpoint:** Navigating directly to `/api/appointments` in a browser renders a color-coded live appointments dashboard with summary stats and a collapsible raw JSON payload, while programmatic requests (`fetch` or `?format=json`) return pure JSON.
8. **No "Testing As" Distractions:** Removed test persona dropdowns. The client app uses the user's phone session with a quiet "Switch number" option.
9. **Doctor Landing & Patient Privacy Isolation:** First-time visitors see Dr. Maya Sen's bio and credentials with a direct "Book an appointment" button. We never default to another client's personal consultations on arrival.
10. **Cancellation Safety Guard:** Protects clients from accidental cancellations with an explicit confirmation dialog before releasing their reserved slot.

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4. What You Added

In One Line:

> **We added an interactive 1-Tap "Still Coming?" pre-session confirmation step that flags unconfirmed bookings as early risks on the dashboard, paired with a dedicated Reminder tracking column and an explicit "Cancelled" status to free slots without corrupting attendance metrics.**

Why:

Nutrition clients ghost due to diet shame, procrastination, or commute friction. Giving them a 1-tap confirmation turns blind no-shows into predictable advance notices, a dedicated reminder column keeps the nutritionist organized with 1-click toast confirmations, and an explicit `Cancelled` status keeps the slot inventory accurate without skewing historical data.

Gaps We Didn’ t Build (Be Upfront About):

- **Client Authentication:** The demo identifies clients by phone number. In production, this requires SMS OTP authentication to prevent unauthorized cancellations.
- **Dashboard Access Control:** The brief asked for a link that opens without login. In production, the practitioner dashboard requires role-based authentication or a PIN.
- **Cancellation Cutoff Rule:** A cancellation 5 minutes before the session is effectively a no-show. A production rule (e.g., free cancellations up to 4 hours before) is needed.

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5. What the AI Got Wrong (and How We Caught and Fixed It)

1. **Unnecessary Marketing Wrapper & Fake Bezels:**
   - *What happened:* The AI initially generated an elaborate presentation wrapper with a simulated iPhone bezel, dynamic island, marketing sidebar, and device toggle badges.
   - *Why it was wrong:* It treated the app like a design mockup rather than a real, usable web application that fills the screen naturally.
   - *The fix:* Stripped away all fake bezels, device toggles, and sidecards. Replaced them with pure CSS responsive layout (single-column max-w-md on mobile; max-w-4xl clean table on desktop).
2. **Color Names in Dropdown Text:**

- *What happened:* The AI placed descriptive color labels in the dropdown text (e.g., `"Booked (Blue)"`).
- *Why it was wrong:* UI labels should be clean and professional; the color should be conveyed strictly through visual CSS pills, not textual explanations.
- *The fix:* Cleaned all options to `Booked`, `Done`, `No-show`, and `Cancelled`.

3. **Slot Deletion on Cancel:**
   - *What happened:* The AI originally deleted cancelled appointments from memory entirely.
   - *The fix:* Introduced the `Cancelled` status so the slot is released on the calendar while maintaining an accurate clinical audit log.

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6. How to Run It

Local Development

```
python server.py
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Open http://localhost:3000 for the Client app, http://localhost:3000/#dashboard for the Nutritionist dashboard, and http://localhost:3000/api/appointments for the live REST API endpoint.

Deployment (Vercel)

1. Push this folder to GitHub.
2. Import the repository into Vercel.
3. `vercel.json` and `api/appointments.js` automatically handle `/dashboard` rewrites and the serverless REST API endpoint with zero configuration.